

# Resource-Normalized Scaling Laws for Byzantine Consensus: Chance-Constrained Validator-Set Sizing from Commodity-CI Measurements

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**Problem.** Sizing a permissioned BFT validator set for national-scale e-governance must reconcile degrading consensus throughput with improving fault-detection power as  $n$  grows; published single-factor exponents are not reproducible across testbeds. **Contributions.** (1) An identifiability theorem:  $\hat{\alpha} \rightarrow \gamma\lambda - \beta$  along a concurrency path of slope  $\lambda$ ; (2) a resource-normalized measurement protocol fixing per-validator CPU quota  $q$ ; (3) coverage-validated lognormal prediction intervals; (4) chance-constrained sizing whose feasible set is a closed interval  $[n_{\min}, n_{\max}]$ . **Validation.** A public CI campaign on one four-core runner covers  $n \leq 16$ , concurrency up to 64, emulated RTT, and omission/duplication fault injection.

**Keywords:** *Byzantine fault tolerance, consensus, scaling law, capacity planning, chance-constrained optimization, e-governance*

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## 1. Introduction

Permissioned Byzantine Fault Tolerant (BFT) consensus underpins e-governance systems — voting, registries, digital identity — and forces a concrete procurement question: *how many validators should the network have?*

The count  $n$  pulls in opposite directions. Throughput argues for small  $n$  (every validator executes every transaction; PBFT-family protocols exchange  $O(n^2)$  messages per block). Safety argues for large  $n$  (detecting a coordinated faulty minority concentrates exponentially in  $n$ ). The two regimes have been studied separately; their joint treatment is the subject of this paper.

Two obstacles intervene. Published power-law fits  $\text{TPS}(n) \propto n^\alpha$  disagree in magnitude and sometimes in sign — our own earlier studies, extrapolated to sixty validators, differ by roughly an order of magnitude — because client concurrency is rarely held fixed while  $n$  varies, so the fitted exponent mixes load-generator scaling with consensus overhead. And large testbeds are unavailable: packing many validators onto one host confounds consensus cost with contention for the host CPU.

**Contributions.** (1) An **identifiability theorem** (Section 3): if throughput follows a bi-factor law and concurrency is scaled along a path of slope  $\lambda$ , the single-factor least-squares fit converges to  $\hat{\alpha} = \gamma\lambda - \beta$ ; a positive exponent is therefore evidence about  $\lambda$ , not about the protocol

(Theorem 1). (2) A **resource-normalized measurement** (RNM) protocol that fixes a CPU quota  $q$  per validator independently of  $n$  and reports  $\widetilde{\text{TPS}} = \text{TPS}/q$ , making the required factorial design feasible on one commodity host;  $q$ -invariance is tested, not assumed. (3) Lognormal **prediction intervals** validated by empirical coverage rather than by unchecked percentage error. (4) Validator-set **sizing as a chance-constrained program** whose feasible set is a closed interval  $[n_{\min}, n_{\max}]$  (Theorem 3); an empty interval certifies that no size meets the requirement. The detection bound itself is Hoeffding’s inequality with a design-effect correction; we claim nothing new for it beyond supplying the opposing constraint with measured parameters. Scope is limited to throughput and detection reliability: cryptography, endpoint security and coercion lie outside, absolute TPS values are environment-specific, and all measurements come from one implementation (Hyperledger Besu/QBFT) on one host; cross-client replication is out of scope (Section 6).

## 2. Related work

*BFT and benchmarking.* Classical PBFT [1] tolerates  $f < n/3$  at  $O(n^2)$  messages [2, 3]; HotStuff [4], Tendermint [5], Algorand [6] and HoneyBadgerBFT [7] vary latency and randomization, while Vukolić [8] argues that no protocol dominates. All fix the validator set at deployment and none supplies a sizing procedure. Empirical characterizations of Fabric [9, 10], BLOCKBENCH [11] and BFT-SMaRt [12] report throughput at fixed configurations without separating offered load from validator count — exactly the confounding of Theorem 1 — and do not extrapolate.

*Scaling, statistics, optimization.* Amdahl [13] and Gustafson [14] bound speedup under a fixed serial fraction; Gunther’s Universal Scalability Law [15] adds a coherency term and is our closest baseline, but does not separate the concurrency factor and has not been calibrated for BFT. Concentration bounds [16, 17], delta-method intervals [19], the residual bootstrap [18] and chance-constrained programming [20, 21] are used as standard tools; what is specific here is the opposing monotonicity of throughput and detection, which turns the feasible set into a closed interval.

*E-governance and prior work.* Remote-voting analyses [22, 23, 24] show that consensus-layer guarantees resolve neither endpoint nor coercion threats. Our earlier single-factor calibration [26] and detection-based adaptive switching [27] are unified and corrected here: we identify the confounding that inflated the exponent of [26] and measure the detector rates that [27] assumed.

## 3. Resource-normalized measurement and the bi-factor law

### 3.1. The RNM protocol

If  $n$  validators share a host of  $Q$  cores, each receives  $Q/n$  cores, so measured throughput mixes growing consensus cost with a shrinking compute budget. An exponent fitted to such data is a property of the host, not of the protocol.

**Definition 1 (Resource-normalized measurement).** *A campaign follows the RNM protocol with quota  $q$  if every validator container receives the same CPU quota and memory limit independently of  $n$ , with aggregate reservation  $nq$  below host capacity. Normalized throughput is  $\text{TPS} := \text{TPS}/q$ .*

**Assumption 1 (Separable compute budget).**  $\text{TPS}(n, c, q) = q \cdot \Phi(n, c) \cdot (1 + O(q))$  on the operating range: the quota enters multiplicatively and does not interact with  $n$  to first order.

The assumption is expected for CPU-bound verification and execution under a cgroup quota; it fails when fixed per-process overhead (e.g. GC) dominates a small  $q$ , when  $nq$  saturates host bandwidth, or when block production is paced by a fixed period. It is a working hypothesis with a stated domain, not a law.

**Statement 1 ( $q$ -invariance).** *Under Assumption 1 and  $\Phi(n, c) = T_0 g(c) n^{-\beta}$ , the OLS slope of  $\log \text{TPS}$  on  $\log n$  at fixed  $c$  does not depend on  $q$ .*

**Proof.**  $\log \widetilde{\text{TPS}} = \log \Phi + O(q)$ ; at fixed  $q$  the  $O(q)$  term is absorbed by the intercept, so the slope equals  $-\beta$  for every  $q$ . ■

Invariance is tested by a partial  $F$ -test on quota  $\times \log n$  interactions across three quota levels (Section 5); if rejected, the exponent is reported only on the surviving quota interval.

### 3.2. The bi-factor law and its identifiability

More submitters  $c$  raise throughput until saturation; more validators  $n$  add communication and verification, lowering it. We model the two multiplicatively.

**Definition 2 (Bi-factor scaling law).** *Under the RNM protocol,*

$$\text{TPS}(n, c) = T_0 \cdot g(c) \cdot n^{-\beta}, \quad g(c) = \frac{c^\gamma}{1 + (c/c^*)^\theta}, \quad (1)$$

with  $T_0 > 0$ ,  $\beta > 0$ ,  $\gamma \in (0, 1]$ ,  $\theta > \gamma$ ,  $c^* > 0$ . In the unsaturated regime  $c \ll c^*$  this is log-linear:  $\log \text{TPS} = \log T_0 + \gamma \log c - \beta \log n$ .

Constraint  $\beta > 0$  encodes replicated-state-machine semantics;  $\theta > \gamma$  makes  $g$  unimodal. Single-factor fits  $\text{TPS} \propto n^\alpha$  can still report  $\alpha > 0$  when the load generator is scaled with the network: the regression attributes the net movement to  $n$  alone.

**Theorem 1 (Identifiability of the consensus exponent).** *Let observations follow the unsaturated bi-factor law with additive noise of mean zero and finite variance, along a concurrency path  $\log c_i = \lambda \log n_i + \mu + \eta_i$  ( $\mathbb{E}[\eta_i | n_i] = 0$ ). If the empirical second moment of  $x_i = \log n_i$  converges to a positive limit, the OLS slope  $\hat{\alpha}_m$  of  $\log \text{TPS}_i$  on  $x_i$  alone satisfies  $\hat{\alpha}_m \xrightarrow{\mathbb{P}} \gamma\lambda - \beta$ . In particular  $\hat{\alpha} > 0 \iff \lambda > \beta/\gamma$ , and  $\hat{\alpha} = -\beta \iff \lambda = 0$ .*

**Proof.** Substituting the path into the log-linear law yields a simple regression with slope  $\gamma\lambda - \beta$  and mean-zero noise. The OLS slope therefore equals that constant plus a ratio  $N_m/D_m$  whose numerator converges to zero in probability (zero-mean finite-variance summands) while the denominator converges to a positive second moment; continuous mapping gives the claim. The sign statements follow from  $\gamma > 0$ . ■

**Remark 1.** The bias is first-order: with  $\gamma = 0.8$ ,  $\beta = 0.5$  and  $\lambda = 1$  one obtains  $\hat{\alpha} = +0.3$ , and extrapolating from  $n = 5$  to  $n = 60$  the two exponents differ by a factor  $\approx 7$ , the order of the discrepancy between our earlier studies [26].

**Corollary 1 (Non-transferability).** *Laboratories measuring along  $\lambda_1 \neq \lambda_2$  obtain exponents differing by  $\gamma(\lambda_1 - \lambda_2)$ ; published exponents are incomparable unless concurrency paths are reported.*

On a factorial grid ( $\lambda = 0$  by construction) we estimate  $(\log T_0, \gamma, \beta)$  by OLS in log-space and  $(c^*, \theta)$  by nonlinear least squares on the full  $c$ -sweep (Section 5).

## 4. Prediction intervals, detection, and the sizing problem

### 4.1. Calibrated prediction intervals

Capacity planning needs a lower bound with a stated probability, not a point estimate. With regressor  $\mathbf{z} = (1, \log c, -\log n)^\top$  and  $\hat{\vartheta} = (\log T_0, \gamma, \beta)^\top$ , the delta method [19] gives predictive variance  $\hat{V} = \mathbf{z}^\top \widehat{\text{Cov}}(\hat{\vartheta}) \mathbf{z} + \hat{\sigma}^2$ .

**Statement 2 (Lognormal prediction interval).** *An approximate  $(1 - \varepsilon_p)$  interval for  $\text{TPS}(n, c)$  is  $\exp\{\log \widehat{\text{TPS}} \mp z_{1-\varepsilon_p/2} \hat{V}^{1/2}\}$ ; the one-sided lower endpoint  $\underline{\text{TPS}}_{1-\varepsilon_p}$  uses  $z_{1-\varepsilon_p}$ .*

Because normality of residuals is only approximate, a residual bootstrap [18] supplies empirical quantiles, and coverage is checked on held-out validator counts (Section 5) rather than asserted.

### 4.2. Detection of coordinated omission faults

The bound below is Hoeffding's inequality with a design-effect correction; we claim no novelty for it. Its role is an  $n$ -monotone constraint opposing throughput, with parameters measured rather than postulated.

**Definition 3 (Fault class and detector).** *A subset  $B$  of size  $k$  is faulty if each member fails a proposal round independently with probability  $p_d > p_f$  (omission class; equivocation excluded). The detector flags validator  $i$  when its participation over  $w$  rounds falls below  $\tau \in (p_f, p_d)$ ; detection succeeds when at least one member of  $B$  is flagged.*

**Assumption 2 (Exchangeable correlation).** Participation indicators in  $B$  are exchangeable with pairwise correlation  $\rho \in [0, 1)$ , so the mean carries design effect  $1 + (k - 1)\rho$ .

**Statement 3 (Miss probability and  $n_{\min}$ ).** *With  $k = \lceil \varphi n \rceil$ ,*

$$p_{\text{miss}}(n, w) \leq \exp\left\{-\frac{2wk(p_d - \tau)^2}{1 + (k - 1)\rho}\right\}, \quad (2)$$

*which is nonincreasing in  $n$  and  $w$ . The constraint  $p_{\text{miss}} \leq \varepsilon_s$  holds for all  $n \geq n_{\min}$  with*

$$n_{\min} = \left\lceil \frac{1 + (k - 1)\rho}{\varphi} \cdot \frac{\log(1/\varepsilon_s)}{2w(p_d - \tau)^2} \right\rceil. \quad (3)$$

**Proof.** A miss requires every member of  $B$  to remain unflagged. The effective sample size is  $n_{\text{eff}} = kw/(1 + (k - 1)\rho)$ ; Hoeffding's inequality on the exchangeable mean yields (2), and solving for  $k$  gives (3). ■

### 4.3. Chance-constrained sizing and the feasibility window

**Problem 1 (Validator-set sizing).** Given peak demand  $D_{\text{peak}}$ , levels  $\varepsilon_p, \varepsilon_s$ , window  $w$ , faulty fraction  $\varphi$ , production scale-up  $\kappa > 0$  from RNM units to absolute throughput, and nondecreasing  $\text{Cost}(n)$ , minimize  $\text{Cost}(n)$  subject to  $\mathbb{P}[\kappa \cdot \widehat{\text{TPS}}(n, c^*) \geq D_{\text{peak}}] \geq 1 - \varepsilon_p$  and

$p_{\text{miss}}(n, w) \leq \varepsilon_s$ , where  $c^* = \arg \max g$ . When  $\kappa$  is free, take the smallest value that meets the throughput constraint at  $n = 4$ .

**Lemma 2 (Deterministic equivalent).** *The chance constraint is equivalent to  $h(n) := \log(\kappa T_0 g(c^*)/D_{\text{peak}}) - \beta \log n - z_{1-\varepsilon_p} \hat{V}^{1/2}(n, c^*) \geq 0$ . If  $\hat{V}$  is nondecreasing in  $\log n$  over the extrapolation region, then  $h$  is strictly decreasing there.*

**Proof.** Statement 2 converts the chance constraint into  $\kappa \cdot \widetilde{\text{TPS}}_{1-\varepsilon_p} \geq D_{\text{peak}}$ ; taking logs gives  $h$ . Strict decrease follows from  $\beta > 0$  and the hypothesis on  $\hat{V}$ . ■

**Theorem 3 (Feasibility window).** *Under Lemma 2 and Statement 3, the feasible set is the integer interval  $\{n : n_{\min} \leq n \leq n_{\max}\}$  with  $n_{\min}$  from (3) and  $n_{\max} = \lfloor \sup\{n : h(n) \geq 0\} \rfloor$ . If the set is nonempty and Cost is nondecreasing then  $n^* = n_{\min}$ ; both endpoints cost  $O(\log n_{\max})$  evaluations of  $h$ .*

**Proof.** Detection holds on  $\{n \geq n_{\min}\}$  and throughput on  $\{n \leq n_{\max}\}$  (two opposing half-lines), so the feasible set is an interval. Monotone cost places the optimum at the left endpoint; bisection locates  $n_{\max}$ . ■

**Corollary 2 (Infeasibility certificate).** *If  $n_{\min} > n_{\max}$ , no validator-set size satisfies both requirements: the architecture (sharding, delegation, or a weaker safety target) must change, not merely the provisioning.*

Chance-constrained programming is classical [20, 21]; the contribution is the opposing monotonicity that bounds the feasible set on both sides and certifies infeasibility.

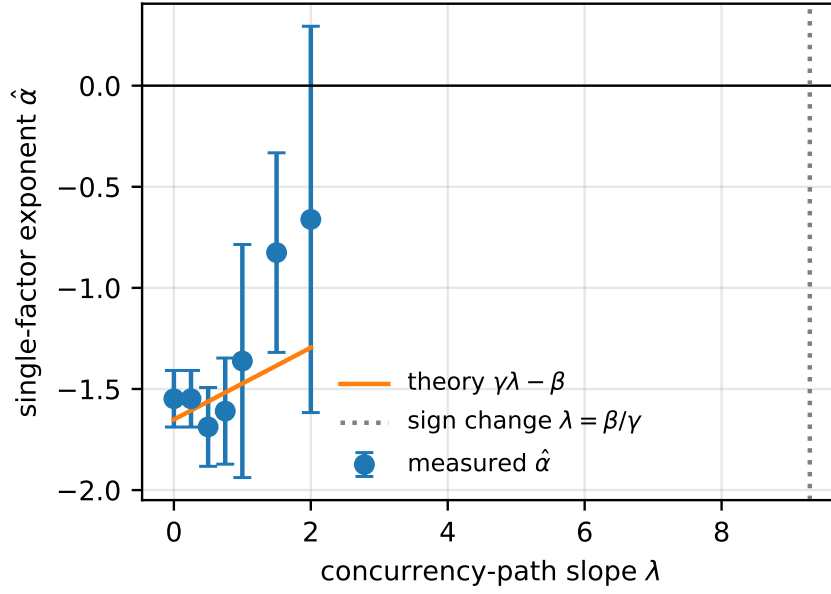
## 5. Experimental validation

### 5.1. Setup and design

The campaign runs in a public continuous-integration workflow on a four-core, 16 GB hosted runner. Each job provisions a Hyperledger Besu [25] QBFT network of  $n$  validators under the RNM protocol (Definition 1); a closed-loop generator drives  $c$  concurrent submitters and records steady-state committed throughput after warm-up.

Faults are injected with kernel traffic control (packet loss and duplication) on designated validators, realizing the omission class of Definition 3 but not equivocation. The detector polls signer metrics over a window of  $w$  rounds to estimate  $p_d$ ,  $p_f$ ,  $\rho$  and the ROC; the Youden threshold is used in Section 6.

Measured groups: **X1** sweeps  $n \in \{4, \dots, 16\}$  at  $c = 16$ ; **X2** tests  $q$ -invariance at three quotas; **X3** sweeps  $c \in \{1, \dots, 64\}$  at  $n = 7$ ; **X4** adds emulated RTT; **X5** injects omission/duplication at faulty fraction  $\varphi = 0.2$ . Cross-client replication on a second QBFT-family implementation (planned as **X9/GoQuorum**) is *out of scope* for this paper: Theorem 1 is a statement about least squares, not about any one client, and the numerical coefficients below are reported as Besu/QBFT-specific by design. Post-hoc: **X3b** re-samples along concurrency paths  $\lambda \in \{0, 0.5, 1, 2\}$ ; **X6** is a discrete-event simulation used only for qualitative extrapolation beyond the measured range (not as a measurement); **X7** predictive coverage on  $n \in \{11, 13, 16\}$ ; **X8** sizing ablation. In total 94 successful records,  $\approx 27.2$  core-hours,  $n_{\max} = 16$ .



**Fig. 1.** X3b: single-factor  $\hat{\alpha}$  along concurrency paths of slope  $\lambda$ , versus  $\gamma\lambda - \beta$  (Theorem 1). Predicted sign change at  $\lambda = \beta/\gamma = 9.29$

**Table 1.** Calibrated parameters, invariance, detector, and coverage

| Quantity                              | Estimate                                      | Note                            |
|---------------------------------------|-----------------------------------------------|---------------------------------|
| $T_0; \beta; \gamma$                  | 2016.5; $1.651 \pm 0.151$ ; $0.178 \pm 0.070$ | bi-factor                       |
| $c^*, \theta; \hat{\sigma}$           | 67.9, 2.542; 0.404                            | saturation / residual           |
| $\hat{\alpha}$ at $\lambda = 0, 1, 2$ | -1.548, -1.362, -0.661                        | X3b                             |
| $\beta$ at $q = 0.20, 0.25, 0.33$     | 1.550, 1.541, 0.879                           | X2, $p = 0.002$                 |
| $\beta$ at RTT 0/high                 | 1.527, 2.708                                  | X4                              |
| $p_d, p_f, \rho; \text{AUC}$          | 0.397, 0.376, 0.000; 0.506                    | X5, $\tau = 0.90$               |
| coverage 90%/95%                      | 90.9 %, 95.5 %                                | hold-out $n \in \{11, 13, 16\}$ |

## 5.2. Results

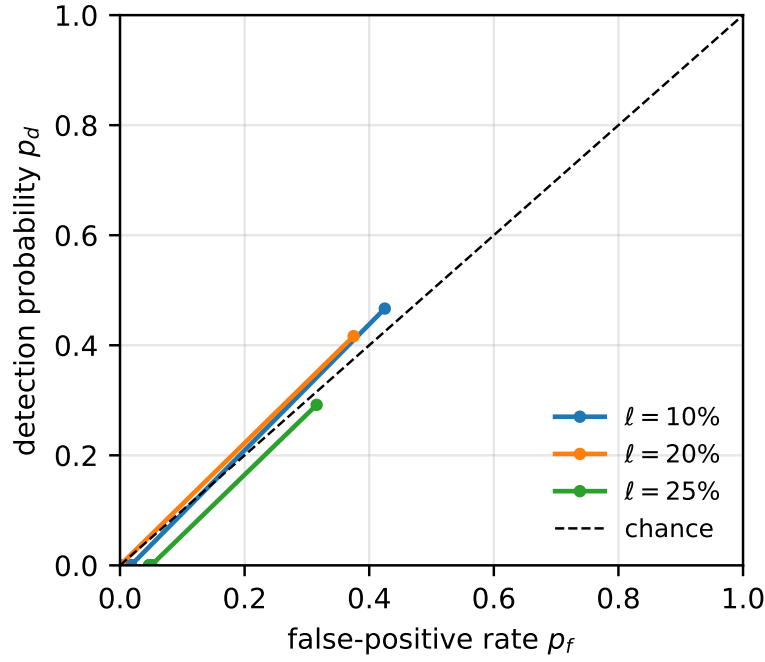
Joint OLS on the reference-quota calibration subset yields  $\beta = 1.651 \pm 0.151$  and  $\gamma = 0.178 \pm 0.070$  ( $R^2 = 0.727$ ); remaining parameters appear in Table 1. Under RNM one must have  $\beta > 0$ ; a single-factor fit along  $\lambda = 1$  instead gives  $\hat{\alpha} = -1.362$  against the predicted  $\gamma\lambda - \beta = -1.473$ .

Figure 1 confirms that estimated exponents track the predicted line. The partial  $F$ -test for quota  $\times \log n$  interaction rejects exact  $q$ -invariance ( $p = 0.002$ ), so coefficients are reported at the reference quota  $q = 0.25$  only;  $\beta$  rises from 1.527 at zero RTT to 2.708 under elevated latency (X4).

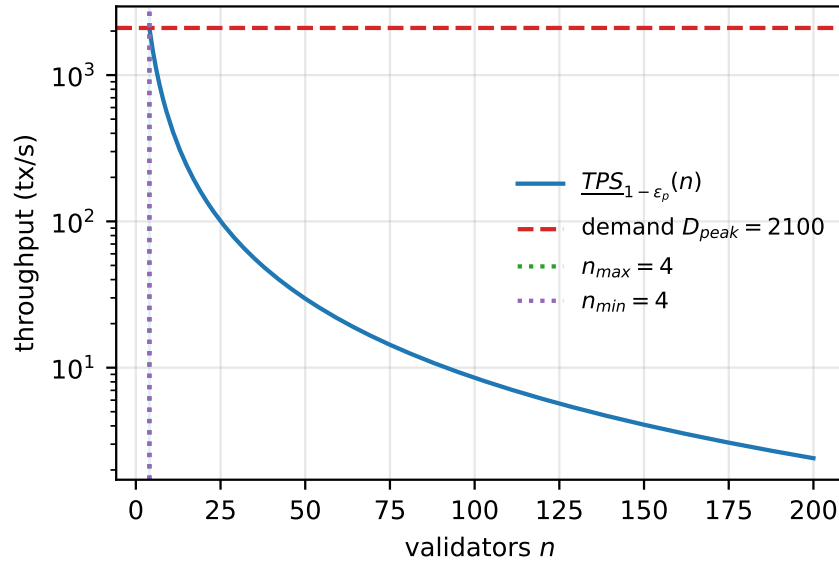
Figure 2 shows the weak detector (AUC 0.506). Hold-out coverage meets the nominal levels in Table 1. Out-of-sample log-RMSE for linear / single-factor / USL [15] / bi-factor predictors is 1.329, 0.523, 0.637, 0.538; relative sizing costs are 4.00, 1.50, 1.50, 1.00 (Section 6).

## 6. Application to an e-governance workload, and limitations

We instantiate Problem 1 on a national-scale profile of about  $3 \times 10^7$  ballots over twelve hours (peak demand  $D_{\text{peak}} \approx 2.1 \times 10^3$  TPS) with the requirement that a coordinated omission fault of fraction  $\varphi = 0.2$  be detected within one epoch at level  $1 - \varepsilon_s$ .



**Fig. 2.** X5 detector ROC for omission/duplication faults. AUC = 0.506 at the Youden threshold  $\tau = 0.90$ ; safety is consequently weak and the sizing window is set by throughput



**Fig. 3.** Feasibility window at minimal admissible  $\kappa = 13.00$ : confidence-adjusted absolute throughput versus  $D_{\text{peak}}$ ; shaded band  $[n_{\min}, n_{\max}] = [4, 4]$

Because the law is calibrated on RNM-normalized throughput, absolute demand enters through an explicit scale-up  $\kappa$ . Taking the *minimal admissible*  $\kappa$  (smallest value whose lower prediction bound at  $n = 4$  meets  $D_{\text{peak}}$ ) yields  $\kappa = 13.00$ , operating concurrency  $c^* = 32$ , and feasibility window  $[4, 4]$  with  $n^* = 4$  (Theorem 3); Figure 3 shows the intersection. Detector rates enter at the Youden threshold  $\tau = 0.90$  of X5.

The window is most sensitive to  $\beta$  (shifts  $\log n_{\max}$ ), to  $\rho$  (inflates  $n_{\min}$  via (3)), to RTT-elevated  $\beta$  from X4, and to tightening  $\varepsilon_p$  or  $\varepsilon_s$  (closing from both sides). Operationally: with confidence  $1 - \varepsilon_p$  the network sustains the peak up to  $n_{\max}$ , detects the specified fault from  $n_{\min}$  upward, and the cheapest admissible configuration is  $n^*$  at production scale  $\kappa$ .

**Remark 2.** This illustrates the sizing workflow; it is not a recommendation to run binding elections on a blockchain [22, 23, 24]. The same workflow applies to registry and identity workloads.

## 6.1. Limitations

*Fault class.* Encrypted transport precludes selective consensus-message injection, so X5 realizes omission and duplication only. Signer-metrics detection is weak (AUC 0.506), the safety constraint is non-binding at the operating point, and the window is set by throughput.

*Absolute TPS and range.* RNM quotas keep absolute throughput far below dedicated-hardware benchmarks; every claim concerns exponents, coverage and sizing. Absolute demand enters only through  $\kappa$ . The largest measured set is 16; beyond it, intervals of Statement 2 quantify but do not eliminate extrapolation risk. With  $\beta = 1.651$  the window collapses to  $n = 4$  — a substantive conclusion, not an artefact.

*Single host and platform.* Theorem 1 is about least squares and is platform-independent; the calibrated  $(\beta, \gamma)$  are not. Cross-client replication (GoQuorum/X9) is deliberately out of scope, so all numerical coefficients are Besu/QBFT-specific. Exact  $q$ -invariance is rejected on  $\{0.20, 0.25, 0.33\}$ , and the fit is reported at  $q = 0.25$  only. Structural claims (bias form, feasible-set shape) are the durable part; numerical values are provisional.

## 7. Conclusions

We addressed validator-set sizing for permissioned BFT when no large testbed is available. The identifiability theorem shows that a single-factor exponent converges to  $\gamma\lambda - \beta$ , so a positive fit is consistent with a load generator scaled alongside the network and explains the discrepancy between our earlier predictions [26, 27]. The RNM protocol makes the required factorial design usable on one commodity host; where exact  $q$ -invariance fails, the law is reported at the reference quota. Coverage-validated prediction intervals replace unchecked point accuracy, and coupling the throughput law with a Hoeffding-type detection bound yields a chance-constrained program whose feasible set is the closed interval  $[n_{\min}, n_{\max}]$ , with absolute demand entering through  $\kappa$ . Of the four results, the first — a statement about estimation — is the most likely to outlive the specific measurements. The full campaign runs in a public CI workflow on one four-core runner.

**Future work.** Equivocation-capable instrumentation; multi-client validation; latency/finality constraints in the window; online recalibration of  $\beta$ .

## Conflict of interest

The authors declare no conflict of interest.

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## Data availability

Workflow definition, generators, fault-injection and analysis code, and raw artefacts are available at [https://github.com/bpenyak/paper\\_3](https://github.com/bpenyak/paper_3).

## Use of artificial intelligence

AI-assisted tools were used for language editing and typesetting. All scientific results, proofs, design and analysis were produced by the authors.

## Contributions of authors

**Peniak B. O.:** conceptualization, methodology, formal analysis, software, investigation, data curation, writing — original draft, visualization. **Liubinskyj B. B.:** supervision, validation, writing — review and editing.

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## РЕСУРСНО-НОРМОВАНІ ЗАКОНИ МАСШТАБУВАННЯ ВFT-КОНСЕНСУСУ: ІМОВІРНІСНО-ОБМЕЖЕНЕ ВИЗНАЧЕННЯ РОЗМІРУ МНОЖИНИ ВАЛІДАТОРІВ ЗА ВИМІРЮВАННЯМИ НА ТИПОВІЙ СІ-ІНФРАСТРУКТУРІ

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Метод визначення розміру множини валідаторів ВFT-мережі для електронного урядування: пропускна здатність спадає зі зростанням  $n$ , а потужність виявлення відмов зростає з  $n$ . Результати: 1) протокол ресурсно-нормованих вимірювань із фіксованою CPU-квотою  $q$ ; 2) двофакторний закон і теорема ідентифікованості ( $\hat{\alpha} = \gamma\lambda - \beta$ ); 3) лог-нормальні прогнозні інтервали з перевіркою покриття; 4) задача з імовірнісними обмеженнями та теорема про вікно  $[n_{\min}, n_{\max}]$ . Експерименти — у публічному СІ на одному чотириядерному вузлі.

**Ключові слова:** візантійська відмовостійкість, консенсус, закон масштабування, планування потужності, оптимізація з імовірнісними обмеженнями, електронне урядування

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